

Topic P6

Space physics

Concept guide · Worked examples · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, common misconceptions, worked applications and exam-style practice for this topic.

Syllabus map

P6.1	Earth and the Solar System
P6.2	Stars and the Universe

How to study: retrieve definitions and equations, cover the worked examples, then complete the practice paper without using the notes.

1. The Solar System and gravity

The Solar System contains the Sun, eight planets, dwarf planets, moons, asteroids and comets. Gravity provides the centripetal force for orbits. Orbital speed depends on orbital radius and period; closer planets generally move faster.

$$v = 2\pi r/T$$

Worked example

A satellite orbits at radius 7.0×10^6 m in 6000 s. $v = 2\pi r/T = 7.3 \times 10^3$ m/s.

Examiner insight

An orbiting object accelerates because its velocity direction changes even if speed is constant.

2. The Sun and energy

The Sun is a medium-sized star containing most of the Solar System mass. Fusion of hydrogen nuclei releases energy. Electromagnetic radiation travels from the Sun through the vacuum; light takes about 8 minutes to reach Earth.

Examiner insight

Solar wind is particles; sunlight is electromagnetic radiation.

3. Stellar formation and stable stars

Stars form when gravity contracts gas and dust in a nebula. Temperature rises until fusion begins. During the stable main-sequence stage, inward gravitational force balances outward pressure produced by high-temperature gas and radiation.

Examiner insight

The balance in a stable star is gravitational collapse inward versus pressure outward.

4. Stellar evolution

A Sun-like star becomes red giant, planetary nebula and white dwarf. A massive star becomes red supergiant and explodes as a supernova, leaving a neutron star or, for the greatest masses, a black hole. Supernovae form and distribute heavy elements.

Examiner insight

Initial mass determines the path after the main-sequence stage.

5. Galaxies and cosmological distances

A light-year is the distance light travels in one year, not a unit of time. The Milky Way is about 100 000 light-years across and is one of billions of galaxies. Looking far away means looking back in time because light takes time to arrive.

distance = speed × time

Worked example

Light travelling for 4.0 years covers 4.0 light-years.

Examiner insight

Keep light-year as a distance unit.

6. Redshift and the expanding Universe

Light from distant galaxies is shifted to longer wavelengths. Greater redshift generally corresponds to greater recession speed. Hubble observations support an expanding Universe and imply a finite age from a hotter, denser early state.

Examiner insight

Redshift does not mean that red light alone was emitted; the whole spectrum is displaced.

Exam technique

High-value habits

Use precise definitions, show every calculation step, include units, apply concepts to unfamiliar situations, and answer the command word directly.

Retrieval checklist

- Recall all definitions and equations without notes.
- Explain one unfamiliar application using a linked chain of reasoning.
- Identify the first error in each incorrect response.
- Complete one timed set and mark it only after finishing.

Practice Paper

Topic P6

pace physics**Instructions**

Answer all questions. Select one option for multiple choice. Show working in open spaces and write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A · Multiple choice

Choose the best answer for each question.

1. What provides the centripetal force for a planet orbiting the Sun? [1]
 - A gravity
 - B magnetism
 - C air resistance
 - D nuclear force

2. A planet completes an orbit of radius r in period T . Its orbital speed is [1]
 - A r/T
 - B $2\pi r/T$
 - C $T/2\pi r$
 - D $2\pi T/r$

3. Why is an orbiting satellite accelerating at constant speed? [1]
 - A its mass changes
 - B its direction changes
 - C gravity is zero
 - D its period is infinite

4. What process releases energy in the Sun? [1]
 - A fission of uranium
 - B fusion of hydrogen
 - C chemical combustion
 - D radioactive beta decay only

5. Where do stars begin to form? [1]
 - A nebulae
 - B planetary cores
 - C empty vacuum
 - D black-hole event horizons

6. What balances gravity in a stable main-sequence star? [1]
 - A outward pressure
 - B friction
 - C magnetic attraction
 - D planetary orbits

7. What is the final remnant of a Sun-like star? [1]
 - A black hole
 - B neutron star
 - C white dwarf
 - D red supergiant

8. What event forms and disperses many heavy elements? [1]
- A lunar eclipse
 - B supernova
 - C planetary orbit
 - D radio transmission
9. A light-year is a unit of [1]
- A time
 - B speed
 - C distance
 - D power
10. Redshift from distant galaxies supports the idea that the Universe is [1]
- A contracting
 - B expanding
 - C unchanging and static
 - D inside the Solar System

Section B · Structured questions

11 A satellite moves in a circular orbit of radius 8.0×10^6 m with a period of 7200 s.

(a) Calculate its orbital speed. [3]

(b) State the direction of the resultant force. [1]

(c) Explain why the satellite accelerates although its speed is constant. [2]

(d) State what provides the force. [2]

[Total: 8]

12 The mean Earth-Sun distance is 1.5×10^{11} m. Light speed is 3.0×10^8 m/s.

(a) Calculate the travel time of sunlight to Earth. [2]

(b) Convert the answer to minutes. [1]

(c) State the process that releases energy in the Sun. [1]

(d) Explain why fusion requires very high temperature and pressure. [4]

[Total: 8]

13 A star forms from a nebula.

(a) Describe how gravity leads to formation of a protostar. [2]

(b) State what begins when the core becomes hot enough. [1]

(c) Explain the two opposing effects balanced in a stable star. [3]

(d) State two factors that determine a star's luminosity. [2]

[Total: 8]

14 Compare the evolution of a Sun-like star and a very massive star.

(a) State the stage both enter after the main sequence. [1]

(b) Give the final stages for the Sun-like star. [3]

(c) Give the final stages for the very massive star. [3]

(d) State why supernovae are important for planets and life. [1]

[Total: 8]

15 A galaxy is 2.0×10^8 light-years away.

(a) Explain what a light-year measures. [2]

(b) State how long the observed light has travelled. [1]

(c) Explain what redshift indicates about the galaxy. [2]

(d) Describe how observations of many redshifts support an expanding Universe. [3]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Award marks for correct physics and method, not only the final value. Mark each part separately.

Solutions · Multiple choice

1 A

2 B

3 B

4 B

5 A

6 A

7 C

8 B

9 C

10 B

Solutions · Structured questions

11. 11(a) $v = 2\pi r/T = 2\pi(8.0 \times 10^6)/7200 = 7.0 \times 10^3$ m/s [3]. (b) Towards the centre [1]. (c) Velocity direction changes, so acceleration exists [2]. (d) Gravitational attraction [2].
12. 12(a) $t = d/v = 1.5 \times 10^{11}/3.0 \times 10^8 = 500$ s [2]. (b) 8.3 min [1]. (c) Nuclear fusion [1]. (d) High temperature gives nuclei kinetic energy; high pressure/density increases collision rate; both help overcome electrostatic repulsion [4].
13. 13(a) Gravity contracts gas/dust and temperature rises [2]. (b) Fusion [1]. (c) Inward gravity balances outward gas/radiation pressure [3]. (d) Surface temperature and size/radius [2].
14. 14(a) Red giant/supergiant [1]. (b) Red giant → planetary nebula → white dwarf [3]. (c) Red supergiant → supernova → black hole [3]. (d) Heavy elements are formed/dispersed [1].
15. 15(a) Distance travelled by light in one year [2]. (b) 2.0×10^8 years [1]. (c) Wavelengths shifted longer; galaxy recedes [2]. (d) Most galaxies recede and more distant galaxies show greater redshift/recession, so space is expanding [3].

Self-assessment

Circle the first incorrect step and classify it as knowledge, equation choice, rearrangement, unit conversion, graph/diagram interpretation or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.